

Lithium isotope enrichment for fusion power: challenges and opportunities for the UK

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ABSTRACT

Increasing demand for electricity, net zero targets, and concerns about energy independence and security mean that the UK Government is looking to turn the UK's world-leading nuclear fusion research into an industry that can provide a low-carbon, reliable source of energy in the future.

Fusion power plants are fuelled by breeder blankets containing enriched lithium-6. At present, the UK lacks a process-scale lithium-6 enrichment capability, and without it, it cannot produce the fuel required for a burgeoning fusion energy industry.

This paper discusses the challenges in the supply chain, regulation, and security of enriched lithium-6, including the need to expand regulatory capability and capacity and to clarify and develop export controls. Achieving this capability will require clear policy decisions and leadership from the UK Government, but it has the potential to yield benefits for the nuclear fusion and fission sectors, the developing British lithium mining industry, and the UK's reputation as a global leader in securing and regulating enriched lithium-6.

1. Introduction

The UK's demand for electricity is expected to double between 2023 and 2050, so the Government is looking to nuclear fusion as a low-carbon, reliable source of energy that can contribute to the National Grid's baseload (DESNZ, 2023). This mirrors a broader global trend towards significant public and private investment in the research and development of nuclear fusion for energy production (BEIS, 2021a; Hsu, 2023; IAEA, 2024): the Fusion Industry Association's 2025 annual report stated that there are now fusion companies headquartered in more than a dozen countries, and that many governments have enacted policies to support fusion energy programmes.

In 2021, the UK Government published *Towards Fusion Energy: The UK Government's Fusion Strategy*, followed in 2023 by the UK Government's document *Powering Up Britain*, delivering a blueprint for how the UK will transition to a net-zero economy by 2050 while increasing energy independence and security (HM Government, 2023). In the *Spending Review 2025*, £2.5 billion was allocated for the development of nuclear fusion (HM Treasury, 2025).

These policy decisions build upon a strong tradition of fusion research, which began in the UK in the late 1940s. The country's most

significant fusion experiments to date, the Joint European Torus (JET) and Mega Ampere Spherical Tokamak (MAST), are both located at the Culham Centre for Fusion Energy (CCFE) in Oxfordshire, the headquarters of the United Kingdom Atomic Energy Authority (UKAEA), which has been home to the UK's fusion programme since 1965. (Culham Centre for Fusion Energy)

Looking to the future, the UK Government is investing in the Spherical Tokamak for Energy Production (STEP) programme, a prototype fusion energy plant located on the site of a former coal power plant in Nottinghamshire, which is scheduled for delivery by 2040. It is hoped that STEP will demonstrate that fusion energy is commercially viable and generate thousands of new jobs in the area (DESNZ, 2023; BEIS, 2021a; HM Treasury, 2025).

STEP, like the majority of proposed fusion reactors, will generate energy by fusing deuterium and tritium, two isotopes of hydrogen. It is not viable to directly fuel the reactors with tritium, however, due to supply shortages. The only commercial source of tritium is heavy water reactors, and according to UKAEA estimates, the global stockpile will not be sufficient to fuel the experimental fusion reactors currently under development, let alone a fleet of power plants (Coleman and Kovari, 2018).

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The proposed solution is to generate tritium in the reactor from lithium. Lithium has two naturally occurring, stable isotopes; 92.5% of natural lithium is lithium-7, and the remaining 7.5% is lithium-6. A process known as enrichment, which increases the proportion of lithium-6, will be essential for this future generation of fusion reactors. The lithium, now enriched in lithium-6, is fabricated into a 'breeder blanket', which is installed in a fusion reactor, where it is used to generate the tritium needed to fuel the fusion reaction.

Lithium-6 has a higher neutron cross-section than lithium-7, so it absorbs neutrons much more readily, resulting in more fission reactions, which produce more tritium (Murali et al., 2021). Increasing the proportion of lithium-6 via enrichment, therefore, helps sustain the fusion reactions that power the plant. (Zhen and Todd) The reactions produce heat, which is extracted via a coolant fluid which passes through a steam generator, which drives a turbine and produces electricity (Kembleton et al., 2025).

The tritium breeding ratio (TBR) is the ratio of tritium produced in a breeder blanket to the amount consumed in the plasma. To sustain a fusion reaction, this must be at least one, and designs typically strive to achieve 1.05 or higher to allow for losses. TBR is sensitive to the breeder blanket composition, for which there are several proposed materials, as well as the lithium-6 enrichment level (Zhen and Todd; Abdou et al., 2015; Boccaccini et al., 2022).

Enrichment of lithium-6 is therefore one of the steps needed to make breeder blankets for fusion power plants. This whole process is shown in Fig. 1.

Industrial-scale production of lithium-6 was carried out using the COLEX (COlumn EXchange) process at the Y-12 Plant National Security Complex in Oak Ridge, Tennessee, USA during the 1950s and 1960s to supply their nuclear weapons programme (Giegerich et al., 2019). The COLEX process is now banned in the USA and is unlikely to receive regulatory approval in the UK, as it requires significant quantities of liquid mercury, use of which is restricted by the Minamata Convention of 2013 (Giegerich et al., 2019; Minamata Convention on Mercury; World Nuclear Association, 2024).

The current modest global demand for lithium-6 is most likely met by material originating from the stockpile produced at Y-12; however, this will be insufficient to fuel a future generation of fusion reactors (Murali et al., 2021). At the time of writing, Russia (and possibly China, although this is unclear) is the only country actively producing lithium-6, and this is not available on the open market (World Nuclear Association, 2024; CCFE, 2019). Russia achieves this large-scale production using an electrolysis process with a mercury cathode (World Nuclear Association, 2024). Lithium isotope separation can be achieved at a laboratory scale using alternative separation methods; however, this is not done at a process scale anywhere in the world (CCFE, 2019).

It is challenging to predict the amount of lithium-6 that a global fusion industry would require. 112 kg of lithium-6 are consumed per full-power year per GW of energy generated (Giegerich et al., 2019). This number, however, is small compared to the total inventory required. Inventories will vary based on the power output, breeder blanket material, and enrichment level required, and it is too soon to predict how the number of fusion power plants will grow over time (Magaud et al., 2004).

As an example, the 2 GW DEMO fusion power plant, utilising a water-cooled lithium lead blanket with lithium-6 enriched to 90%, requires 52 tons of pure lithium-6 (Giegerich et al., 2019). However, Ward et al. have stated that approximately 10,000 GWth-scale fusion power plants would be required to meet 20% of current global primary energy demand (Ward et al., 2025). In a proposed scenario where 10 plants are operational in 2050, over 100 tonnes of lithium-6 would be required by 2035. The number of fusion power plants, and consequently the demand for lithium-6, is then anticipated to grow tenfold every decade to meet the target of 10,000 fusion power plants. The UK supplied 1.1% of the world's energy in 2024 (Ersoy and Schaffer, 2025). Assuming that this proportion remains constant and the 10,000 fusion power plants are

distributed accordingly, the UK would host 110 of them. These plants would have a total inventory of 2860 tons of pure lithium-6 (based on DEMO requirements) and an annual consumption of >12 tonnes of lithium-6, assuming 100% availability of output power. This scenario is based on a number of assumptions; hence, it is best used as an indication of the scale of lithium-6 demand in the UK.

The current absence of a lithium-6 supply chain limits the potential for the global fusion industry to grow, a reality that has begun to be acknowledged in recent years. UKAEA launched the Fusion Industry Programme (FIP) challenge "Realising the potential of lithium in an economic, sustainable and scalable fusion energy fuel-cycle" to encourage research institutions and businesses to develop fusion fuel technologies (UKAEA, et al., 2023). Between 2023 and 2025, this programme provided funding to project teams researching four different lithium-6 enrichment technologies:

- Solvent extraction using crown ethers (University of Manchester);
- Microbial separation (Bangor University);
- Solution centrifugation (University of Bristol);
- Plasma-based separation (Frazer-Nash).

In 2024, the German Government provided a research grant to a collaborative project working at the Karlsruhe Institute of Technology to develop a mercury amalgam process that is cleaner and more efficient than COLEX (Giegerich et al., 2019; Fusioneering, 2024), and a demonstration plant in the US for a laser-based method was completed in 2025 (Power Technology, 2025). If any of these methods are scaled up to the process level and commercialised, the plants will be the first of a kind (FOAK).

1.1. Developing lithium-6 enrichment capability in the UK

One of the aims of the UK Government's fusion strategy is to turn the world-leading fusion research into a world-leading fusion industry (BEIS, 2021a). Similarly, the UK Government's document *Powering Up Britain* emphasises the need for energy security (HM Government, 2023). These two objectives can only be met if the UK has a process-scale domestic lithium-6 enrichment capability.

This paper will highlight the challenges and opportunities associated with delivering an industrial-scale process to enrich lithium in the UK. Moreover, this paper will also demonstrate how investment in lithium-6 enrichment can yield broader benefits and provide recommendations for addressing the associated challenges.

2. Discussion

2.1. The fusion technology supply chain

The Fusion Industry Association (FIA) document *The Fusion Industry Supply Chain: Opportunities and Challenges* gives a global recommendation on the three things that are necessary for the commercialisation of any fusion reactor design:

- Fuel to power the reactor.
- A skilled workforce to produce and assemble the reactors.
- Specialist, highly engineered materials and components (Ersoy and Schaffer, 2025).

Lithium-6 enrichment, as a crucial component of fuel production, is a key part of the supply chain. Moreover, the opportunities and challenges associated with lithium-6 enrichment are not isolated; they influence

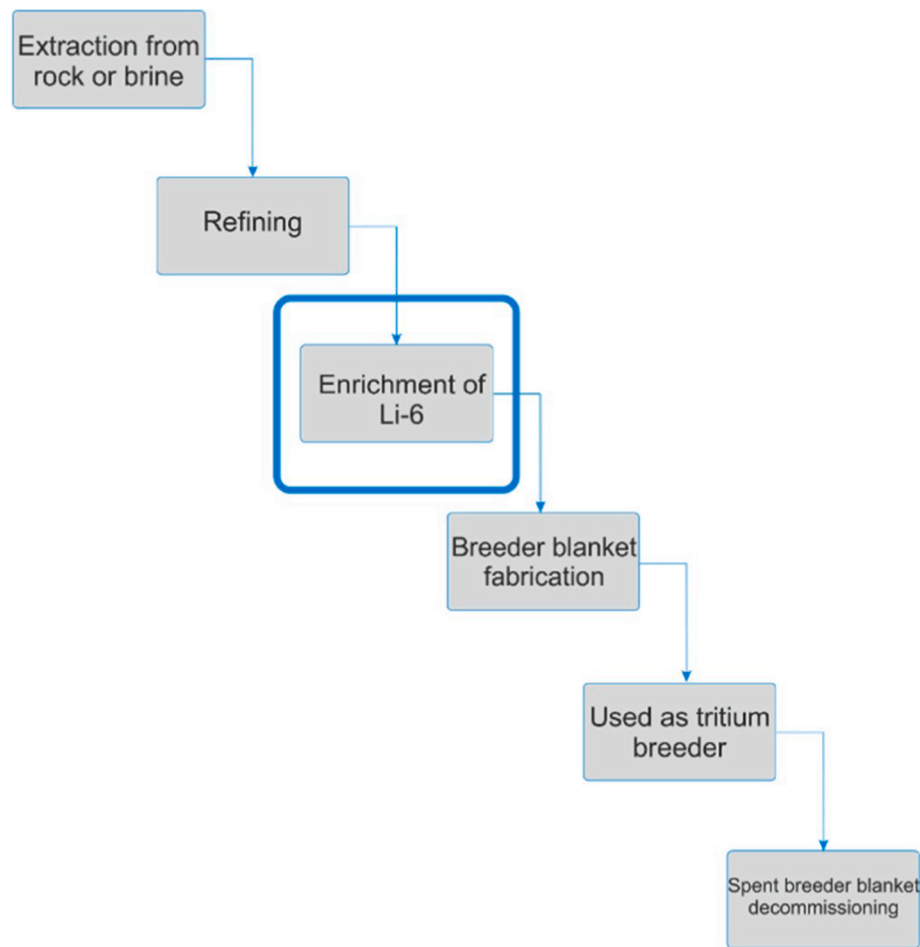


Fig. 1. Lithium-6 will be mined, refined, converted into a breeder blanket, and then used to power fusion reactors.

other segments and shape the entire supply chain.¹

The Fusion Industry Supply Chain: Opportunities and Challenges identifies potential problems the industry may face when establishing this supply chain. While some of these challenges relate to the availability of raw materials and specialist equipment, there is one challenge which could be more readily resolved with government support: for fusion to scale up from research and development to being a key component of the global energy infrastructure, suppliers need to expand their capacity; something they are currently hesitant to do due to the uncertainty around the timescales of deploying fusion. However, fusion energy **cannot be deployed at scale without that supply chain being in place, creating a chicken-or-egg scenario** (FIA, 2023). The FIA concluded that the fusion energy supply chain and the broader fusion industry must **grow in parallel**, rather than one preceding the other.

The FIA also discusses the need for risk-sharing mechanisms and suggests that national laboratories collaborate with private companies, rather than competing with them (FIA, 2023). They highlight the success of a public-private partnership in the US, where early investment in microchip technology by the Department of Defence helped to create the modern technology ecosystem (FIA, 2023; FIA, 2025). The lithium-6 enrichment FIP challenge represented a successful collaboration between UKAEA, several universities, and a private enterprise; however, the funding was limited to 18 months, and there is now a reliance on other sources, such as private companies, for follow-on investment. If

these lithium-6 enrichment technologies continue to develop and are scaled up and spun out into companies, support and collaboration must continue as the technology advances, bridging the gap between early investment in research and a successful commercial enterprise.

The UK Government announced in June 2025 that it would invest £2.5 billion in nuclear fusion over five years (HM Treasury, 2025). £1.3 billion of this is earmarked for STEP, and £180 million for building a facility for the Lithium Breeding Tritium Innovation Programme (LIBRTI), which is working to develop fusion fuel technology for power plants (DESNZ, 2026). An additional £110 million has been allocated for support to the wider industry and the supply chain. While longer-term funding provides confidence to suppliers as they attempt to scale their operations, the FIA also recommends measures such as grants and loans, tax credits, market guarantees, and fixed-fee tariffs to support investment in the fusion supply chain (FIA, 2025).

The common thread here is that **uncertainty can prevent or delay investment in the supply chain**. While the measures recommended by the FIA can mitigate the risks to investors, concrete decisions and commitments from the Department for Energy Security and Net Zero (DESNZ), in addition to the recent funding, can help to provide the certainty needed for market growth in the supply chain.

2.2. Regulation of lithium-6 enrichment technology

The Energy Act 2023 made the UK the first country to write fusion-specific legislation into law (DESNZ, 2024a; Energy Act 2023, 2023, Parliament). It excluded nuclear fusion plants from the regulatory and licensing requirements of the Nuclear Installations Act 1965, meaning that they would not be considered nuclear licensed sites and would

¹ For more detailed information on the fusion supply chain, the FIA reports are recommended reading: *The Fusion Industry Supply Chain: Opportunities and Challenges 2023* and *The Fusion Industry Supply Chain 2025*.

therefore not fall under the jurisdiction of the Office for Nuclear Regulation (ONR). Fusion research and development is currently regulated by the Environment Agency (EA) and the Health and Safety Executive (HSE), who oversee the environmental and the health and safety aspects of fusion activity, respectively, and will continue to do so as the industry expands (BEIS and DESNZ, 2023; BEIS, 2022). At present, however, there is no regulation specifically for other aspects of fusion, such as licensing designs or export controls (BEIS and DESNZ, 2023; BEIS, 2022; RHC, 2021).

The novel status of process-scale lithium-6 enrichment techniques means the UK Government has not previously been required to regulate them. As both lithium isotopes are stable and there will be no radiological materials present, lithium-6 enrichment will not automatically fall under the fusion energy regulatory regime.² Instead, the key pieces of legislation governing the health, safety and environmental aspects of fusion fuel production are summarised in Table 1. The safeguarding, security and export control regulatory requirements are discussed in section 2.3 “Security, safeguards, and export control”.

2.2.1. Regulatory capability and capacity

The EA and the HSE are experienced in overseeing the environmental and health and safety aspects of fusion research and development projects in the UK and have regulated hazardous industrial processes under the COMAH regulations for many decades (RHC, 2021; BEIS, 2021b). However, as the primary regulators for a commercial fusion energy industry, they will be required to oversee substantially more facilities, and their remit will expand to areas they may not currently cover. The Department for Business, Energy and Industrial Strategy document *Towards Fusion Energy: The UK Government's response to the consultation on its proposals for a regulatory framework for fusion energy* states that regulators need to build their capability and capacity over the coming decades to regulate fusion energy facilities effectively (BEIS, 2022).

This is supported by the Regulatory Horizons Council (RHC), an independent committee that advises the UK Government on the regulatory reform required to support technological innovation. The RHC suggests that the EA and the HSE internally assess any gaps in their skill sets for managing different fusion technologies and upskill as appropriate. The report also states that the EA is aware of the need to build its capability for regulating a future large-scale fusion power plant (RHC, 2021). However, neither the EA nor the HSE's 2025–2026 business plans refer to fusion, and only the EA's business plan mentions nuclear at all (Environment Agency, 2025; HSE, 2024).

There is therefore recognition that the EA and the HSE need to expand their capability and capacity to handle the more extensive regulatory responsibilities they will hold as fusion energy is commercial-

Table 1

Key health, safety and environmental regulations for the manufacture of fusion fuel in England. Adapted from *Towards Fusion Energy: The UK Government's proposals for a regulatory framework for fusion energy* (BEIS, 2021b).

Regulatory body	Regulations/legislation
Environment Agency	Environmental Permitting (England and Wales) Regulations 2016
Health and Safety Executive	Health and Safety at Work etc. Act 1974 Control of Substances Hazardous to Health 2002
Joint enforcement as the COMAH Competent Authority	Control of Major Accident Hazards Regulations (2015)

² There may be an exception to this if the enrichment plant was co-located on a site that housed other fusion energy facilities which were regulated under the fusion energy regime, such as a reactor, but this would need to be addressed on a case-by-case basis.

ised. This challenge will be compounded by the advent of the UK's SMR programme and the new nuclear fission power plants at Sizewell C and Hinkley Point C, all of which will require regulatory attention (HM Treasury, 2025).

In addition to recruiting staff with relevant experience, this expansion of regulatory capability may include formal training, engagement with organisations such as the UKAEA, universities, or private companies with relevant expertise, as well as international collaboration with other agencies. This would help develop existing personnel's knowledge and ensure that there are people with relevant skills who could join the industry. These efforts may already be underway, but little information is available in the public domain about what regulators are doing to prepare.

2.2.2. Current scope of the regulators' concern

The EA hosted a workshop on early engagement with fusion reactor designers and vendors, as well as DESNZ, in early 2025. The EA has also held meetings with industry representatives from the US, Canada, Japan, South Korea, and the International Atomic Energy Agency (IAEA). However, these events focused on fusion plants and did not address the regulation of any aspects of the fusion supply chain. This was a key theme that emerged during the authors' discussions with both the EA and HSE: work is underway to prepare for commercial fusion power, but only at the power-generating stage, and little attention has been paid to the supply chain so far. This needs to be addressed to avoid future delays and to give suppliers confidence that their chosen processes and designs will obtain regulatory approval, encouraging their investment in the supply chain (FIA, 2025).

Although there is no formal early engagement process for lithium-6 enrichment technologies with the EA and HSE (the ONR has such a process, which enables vendors to receive advice about their reactor designs before they are finalised), (ONRa) contacting the regulator as early as possible during technology development can help reduce the risk that the regulatory process will delay projects. Vendors can initiate informal contact with the regulators, informing them about their work and keeping them abreast of new developments. As their technologies become more mature, they can request to meet more formally with relevant subject-matter experts at the EA and HSE.

2.2.3. The global picture

The US adopted a similar approach to the UK's when it passed the Fusion Energy Act in 2024, which separated fusion energy regulations from nuclear fission regulations (Fusion Energy Act of 2024, 2024). France's Autorité de Sûreté Nucléaire, (the French Authority for Nuclear Safety and Radiation), on the other hand, is responsible for regulating the International Thermonuclear Experimental Reactor (ITER) tokamak facility as well as France's fission plants, nuclear research laboratories, and radioactive waste management sites. (ASN) However, little is publicly available from the international community regarding the regulation of lithium-6 enrichment or breeder blanket manufacturing.

Given the current lack of international leadership, the UK has an opportunity to establish itself as a global leader in fusion supply chain regulation. This would bring benefits to the UK, including:

- International recognition of the strength of the fusion industry in the UK
- Clarity for the fusion supply chain on their regulatory obligations
- The ability to set an international standard that is proportionate and supports innovation
- Establishing a unified global standard, which would facilitate transnational approvals, in contrast to nuclear fission, where regulation has been developed on a country-by-country basis and reactor designs must be modified to align with different regulatory regimes, slowing deployment.

2.3. Security, safeguards, and export control

The section “Regulation of lithium-6 enrichment technology” notes that fusion fuel manufacturing will not automatically fall under the fusion energy regulatory regime and will instead be governed by the legislation outlined in Table 1. Considering the health, safety, and environmental regulations in isolation, however, is too simplistic, as enriched lithium-6 fuel will also be subject to security, safeguarding, and export control restrictions.

The ability of lithium-6 to produce tritium and enhance neutron production means that, in addition to providing the fuel for civil fusion power plants, it can increase the power of thermonuclear weapons. It is therefore listed as a ‘dual-use’ material on the UK Strategic Export Control List (Gov.uk, 2023). There is thus a need for robust security, safeguarding, and export controls to mitigate proliferation risks while enabling its use in peaceful applications, such as commercial fusion technology, ensuring its market potential is realised and that it can be exported.

Additionally, adherence to national and international laws and agreements is vital for maintaining the UK's credibility and standing in the global community. Compliance ensures that the UK meets its obligations under international treaties and contributes to global non-proliferation efforts.

2.3.1. Security

Ensuring the security of lithium-6 enrichment facilities is crucial. In addition to the threat of proliferation, there are the threats of information theft and sabotage. Commercial and technically sensitive information around lithium-6 enrichment is vulnerable to espionage. This may mean that the UK's commercial competitive advantage is lost, or that the information gleaned is used to plan a sabotage event.

In future, if nuclear fusion becomes a key component of the UK energy supply, a lithium-6 enrichment plant may fall under the definition of Critical National Infrastructure (CNI), increasing the likelihood that it is subject to a physical or cyber threat. (National Protective Security Authority) Additionally, fusion energy's potential to contribute to energy security and decarbonisation aligns with the broader goals of supporting the UK's CNI. Thus, disruption to the supply chain, including lithium-6 enrichment, may impact the nation's energy supply, potentially causing further disruption and economic harm. Furthermore, a high-profile physical or cyber-attack on a fusion power plant or associated facility may undermine UK public support for fusion.

Multiple UK regulatory regimes could serve as templates that can be adapted and customised to address the specific security needs and potential threats associated with lithium-6 enrichment facilities and, more broadly, fusion-related facilities. For example, the Office for Nuclear Regulation's (ONR) Fundamental Security Principles could be adapted as they cover the whole fission fuel cycle. (ONRb) By integrating these principles into the security frameworks of fusion facilities, operators can ensure a comprehensive and effective approach to safeguard against unauthorised access, theft, sabotage, and other security threats.

Care must be taken to avoid placing an undue regulatory burden on the industry, facilities, and operators. It is recognised that there may be some resistance to applying security principles from the fission industry to fusion because of concerns that the measures would be disproportionately onerous given the different risk profile associated with fusion technologies. However, the fundamental security principles are universally applicable to regulated industries and critical national infrastructure and can be customised to address the specific security needs and potential threats associated with these technologies.

2.3.2. Safeguards

Safeguarding, as it pertains to materials, involves the use of comprehensive accounting systems to track the movement and use of specific materials of concern. These systems, comprising inventory management, real-time monitoring, and audit trails, provide

transparency and accountability, ensuring that all activities involving this material are documented and monitored, so that any unauthorised use or diversion for proliferation purposes can be identified.

IAEA Safeguards aim to deter the spread of nuclear weapons by the early detection of the misuse or diversion of nuclear material or technology; however, the international safeguards regime does not currently extend to lithium-6 enrichment. (IAEA) This creates a regulatory void that could undermine efforts to control material diversion and misuse, particularly with the expected increase in quantities of the material. The UK Government will need to collaborate closely with key allies to determine whether there is an international need to bring additional aspects of fusion, such as lithium-6 enrichment, under safeguards. If this is necessary, the HSE, EA, and their counterparts in the devolved governments of the UK will need to determine how to manage this reporting and monitoring regime and ensure that the regulators have the necessary resources and capabilities to do so.

In this respect, the UK has an opportunity to lead the development of specific international safeguards for lithium-6 enrichment globally, which may offer a ‘first mover’ regulatory advantage. If the UK establishes a robust framework that benefits its fusion industry, this may influence what other countries do and set an international standard, rather than the UK having to compromise its policies to trade internationally.

2.3.3. Export control

The UK's export controls on lithium-6 are stringent, requiring export licences for any transfer of this material outside the country. According to the UK Strategic Export Control List, controls extend to facilities and equipment used for lithium isotope separation, including specific technologies such as the lithium-mercury amalgam process, ion-exchange systems, and chemical-exchange systems (Gov.uk, 2023; Council Regulation, 2009). This ensures that lithium-6 and related technologies are not diverted to unauthorised uses. These dual-use export controls apply to elemental lithium, alloys, compounds, mixtures containing lithium, manufacturers thereof, and waste or scrap of any of the foregoing when they are enriched to greater than their natural isotopic abundance (approximately 7.5%).

To develop a successful lithium-6 enrichment industry in the UK, it is important to recognise the current disconnect between regulatory classification and strategic risk. While industrial regulation appropriately addresses health, safety, and environmental hazards, it does not address proliferation risk, espionage, or the protection of sensitive technologies. Export controls under the Export Control Act 2002 and associated legislation do apply to lithium-6 and related technologies, but they focus on transfers and end use rather than domestic production, storage, or use. Broader national security legislation, such as the National Security Act 2023 or the National Security and Investment Act 2021, may be engaged in specific circumstances, but it is not designed as a material-specific control regime.

There is, therefore, a gap in the security and safeguards framework for lithium-6 enrichment. The material sits in a grey area where it is treated operationally as an industrial activity, while simultaneously having characteristics that would normally trigger more structured security and non-proliferation considerations.

The UK's existing export control framework must, therefore, evolve to address the dual-use nature and technological complexities of an expanded lithium-6 enrichment industry. Additional control measures could include:

- Embedding systems that can monitor isotopic enrichment levels
- Regular audits
- Robust licensing
- End-use verification
- Coordinating export control measures with global stakeholders to prevent regulatory loopholes and ensure comprehensive oversight.

- Training and guidance for stakeholders to ensure they are aware of their responsibilities under the export control regime and to reduce risks associated with the inadvertent release of sensitive information.

The current export control regulations covering lithium-6 enrichment will need to be revised to address two distinct areas.

1. Enrichment is defined as “greater than its natural isotopic abundance”, which does not distinguish, for example, materials containing 7.6% lithium-6 from those enriched to 90%.
2. The equipment listed applies primarily to lithium-mercury amalgam enrichment processes, which may need to be expanded to cover equipment specific to new enrichment technologies that are developed (Council Regulation, 2009).

2.4. Integration with UK lithium mining

The UK currently imports all the lithium required by the battery, chemical, and metallurgical industries, leaving it vulnerable to disruptions in the mineral supply chain (Survey, 2022; House of Commons Library, 2024). However, the UK has natural reserves of lithium (Survey, 2022; House of Commons Library, 2024), and preliminary work is underway by a number of companies to start extracting it:

- Imerys British Lithium had aimed to produce 21,000 tonnes of lithium carbonate each year from Cornish granite, although this project was placed into a care and maintenance phase in February 2026 (Imerys British Lithium, 2026)
- Cornish Lithium has received approximately £24 million from the UK Infrastructure Bank to try to achieve the commercial extraction of lithium from both granite and mine water.
- Weardale Lithium and Northern Lithium Ltd are both exploring the potential for lithium extraction in the north-east of England

The FIA emphasises that fusion science is a global endeavour but acknowledges that vertical integration offers benefits (FIA, 2023). Mining lithium in the UK for lithium isotope enrichment could increase energy security by mitigating the impact of disruptions to the mineral supply chain and reducing dependence on other countries. The annual output of Imerys British Lithium, if the project proceeded, would meet demand equivalent to 1.3 TW, far exceeding current UK power generation.³

Establishing a lithium enrichment industry around the Cornish mines would reduce the carbon footprint associated with transportation of the raw material and bring benefits to Cornwall, a county that experiences high levels of deprivation, by providing a more consistent baseline income than tourism, which has seasonal fluctuations (Ovenden-Hope et al., 2025).

2.5. Demand and uses for lithium-7

PWR need Li-7

The enrichment of lithium-6 generates a by-product that is enriched in lithium-7. Far from being a waste, lithium-7 in its hydroxide form is crucial within the nuclear fission sector in the operation of Pressurised Water Reactors (PWRs), where it is used to control the pH and reduce corrosion in the primary coolant circuit, and for the manufacture of ion-exchange membranes for PWR coolant water treatment facilities (World Nuclear Association, 2024). Approximately two-thirds of the world's reactors are PWRs (Brown and Greenspan, 2021). It is also used to make

chemical reagents for other nuclear engineering applications, e.g., lubricants (World Nuclear Association, 2024). Very pure (99.95% enriched) lithium-7 is desirable in these applications for the same reason that it is unsuitable for breeder blanket production: its low neutron cross-section reduces tritium production by neutron capture. In that situation, tritium is an unwanted byproduct that increases the hazards and costs associated with decommissioning the reactors.

The reactors under construction at Hinkley Point C and planned for Sizewell C in the UK are UK European Pressurised Reactors (UK EPR) (Environment Agency, 2022). According to the *Reactor Chemistry* chapter of the Generic Design Assessment (GDA), the UK EPR design will incorporate enriched lithium-7, and thus a supply of it will be needed throughout the minimum 60-year lifespans of these reactors (AREVA and EDF, 2012). It has not been possible to determine exactly how much lithium-7 the UK EPR model will require, but a report submitted to the US Government's House of Representatives in 2013 indicated that PWRs typically require approximately 4.6 kg of lithium-7 annually, while the World Nuclear Association (WNA) report that 65 US PWRs use 400 kg of lithium-7 annually, equivalent to 6.2 kg per PWR (World Nuclear Association, 2024; United States Government Accountability Office, 2013).

The UK Government aims to increase its nuclear power capacity to 24 GW by 2050 (DESNZ and Civil Nuclear, 2024). With a capacity of 1.63 GW per reactor (EDF Energy Holdings Ltd, 2024), fifteen UK EPR reactors would need to be built to meet this target. If it is assumed that they require the same amount of lithium-7 as PWRs (4.6–6.2 kg annually), the UK will require approximately 69–93 kg of lithium-7 per year for its reactor fleet. This is not a large amount, but it is critical for reactor operations and the continuity of energy supply.

A DESNZ policy paper published in 2024 described the UK Government's interest in Advanced Modular Reactors (AMRs). These are novel reactor designs that are currently under development. At present, their focus for the UK is on High Temperature Gas Reactors (HTGRs), but other AMR designs, notably Molten Salt Reactors (MSRs), require enriched lithium-7 for their primary coolant loops (DESNZ, 2024b; Electric Power Research Institute, 2020). If the UK builds reactors of these types, more lithium-7 will be required. Globally, current demand for lithium-7 is approximately one tonne per year; however, this is estimated to increase to 250 tonnes per year once MSRs are commercialised, as each reactor requires tens of tonnes in inventory. It must be noted, however, that while MSRs based on FLiBe and FLiNaK would require enriched lithium-7, there are also designs that do not use a lithium salt, for example, ZrF₄-NaF-KF, ZrF₄-KF, or NaF-BeF₂, and it is too early to determine which design, if any, will dominate the market (LeBlanc, 2010; Figueiredo Luiz et al., 2024).

According to the WNA, lithium-7 is produced only in Russia (and possibly in China, although its status is unclear) at present, and up to 80% of the world's lithium-7 originates from the Novosibirsk Chemical Concentrates Plant in Siberia (World Nuclear Association, 2024). In 2025, it was reported that Russia was developing a lithium-7 fluoride production technology, which was an important step towards its ambition to build molten salt reactors (World Nuclear News, 2025). Given current geopolitical tensions, it is inadvisable to rely on this Russian plant for a material that may become increasingly important for the country's energy security. If the UK were to begin producing lithium-7 as a byproduct of lithium-6 enrichment, not only would it supply the needs of the planned domestic reactors, but any excess could be exported.

3. Conclusions and policy implications

One of the aims of the UK Government's fusion strategy is to turn the country's world-leading fusion research into a world-leading fusion industry to help boost energy independence and security and support its net zero targets. This can only be achieved if the UK has a process-scale domestic capability for lithium-6 enrichment.

This report has identified key challenges and opportunities for the

³ 112 kg of lithium-6 is required per year per GW of fusion energy. This would require approximately 1,500 kg of natural lithium to produce, equivalent to nearly 16 tonnes of lithium carbonate. The scenario presented in the introduction for 110 fusion power plants would consume the equivalent of 1760 tonnes of lithium carbonate annually.

UK across the lithium-6 supply chain, regulation, security, safeguards, export controls, and integration with the fledgling UK lithium mining industry. Additionally, the opportunity to develop a lithium-7 enrichment process alongside a lithium-6 enrichment process has been introduced.

While lithium-6 enrichment facilities are expected to be regulated as industrial chemical plants under existing health, safety, and environmental legislation, this classification does not fully capture the strategic implications of the material. Lithium-6 is a dual-use material with clear relevance to tritium production and thermonuclear weapons, yet there is currently no dedicated regulatory framework governing security or non-proliferation within the UK. This creates a gap between how such facilities are regulated in practice and the risks they present.

Key challenges include:

- Regulatory capability and capacity must be expanded through training and recruitment
- A need for early regulatory engagement to prevent delays in licensing FOAK lithium-6 enrichment facilities
- UK Government commitment to supporting a burgeoning lithium-6 enrichment industry

Key opportunities include:

- Establishing a UK lithium-6 enrichment industry, providing jobs, growing the economy, and increasing energy security
- The UK can establish itself as a world leader in regulating fusion supply chains
- The UK can establish itself as a world leader in the development of specific international safeguards for lithium-6 enrichment
- Enrichment of both lithium-6 and lithium-7 can meet the needs of the fission and fusion industries
- Co-siting lithium-6 enrichment plants close to UK lithium mines could bring economic benefits to deprived areas

These opportunities and challenges bring implications not just for policy but also for commercial entities:

- Government policy and funding to provide certainty to the fusion supply chain, thereby encouraging investment
- An increased regulatory and commercial focus on the fusion supply chain, in addition to fusion power plants
- Government policy on producing rather than purchasing lithium-7

Although substantial challenges remain, lithium enrichment offers the UK the opportunity to establish a world-leading production capability and to lead in lithium enrichment regulation. However, clear policy decisions will be needed to provide investors with certainty and confidence, and the development and oversight of the lithium-6 supply chain are vital to the success of the fusion energy industry.

CRediT authorship contribution statement

Lauren J. Dackombe-Rodrigues: Conceptualization, Investigation, Visualization, Writing – original draft. **Philip Hannah:** Conceptualization, Visualization, Writing – original draft. **Nick Tomkinson:** Conceptualization, Investigation, Writing – original draft. **Kathryn George:** Conceptualization, Project administration, Supervision, Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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